The Bianconi-Barabási Model

Theoretical Background

The Barabási-Albert (BA) model explains scale-free networks through **preferential attachment**: a new node is more likely to connect to existing nodes with high degree ($\Pi(k_i) \sim k_i$).

The Bianconi-Barabási model introduces a crucial extension: **Fitness**.

- **Fitness (η):** A non-negative, node-specific property that represents its inherent ability to attract links. It can model talent, quality, or capacity in real-world systems (e.g., the "fitness" of a webpage, a researcher, or a company).
- **Preferential Attachment with Fitness:** The probability that a new node connects to an existing node i is proportional to **both** its degree k_i and its fitness η_i:
  $\Pi(k_i) \sim \eta_i * k_i$
- **The Bose-Einstein Connection:** Bianconi and Barabási showed that the mapping of this network growth process to a quantum gas particle system. In this mapping:
  - Nodes correspond to **energy levels**.
  - Links correspond to **particles**.
  - Fitness maps to an energy ε_i via $\eta_i = \exp(-\beta \varepsilon_i)$.
  - This analogy leads to a groundbreaking prediction: for a distribution of fitnesses $\rho(\eta)$, a **phase transition** can occur. A finite number of nodes (the "condensate") can grab a finite fraction of all links, analogous to how particles accumulate in the ground state in Bose-Einstein Condensation.

Task 1: Implement the Bianconi-Barabási Model

Your goal is to write a function `bianconi_barabasi_graph(N, m, eta_distribution)` that generates a network with `N` nodes.

- **Step 1: Initialization.**
  - Start with a small connected network of `m0 = m` nodes. A star graph with `m+1` nodes is a simple choice.
  - Assign a fitness η to every node from a given distribution `eta_distribution`. A common choice is a uniform distribution in `[0, 1]`.

- **Step 2: Growth.**
    - For each new node `j` (from `m0` to `N−1`):
        1. Assign it a fitness `η_j` from `eta_distribution`.
        2. Add `m` new links from this new node to existing nodes.
- **Step 3: Preferential Attachment with Fitness.**
    - The probability for the new node to connect to an existing node `i` is: `Π(i) = (η_i * k_i) / Σ_j (η_j * k_j)`
    - **Hint:** Use `np.random.choice` with the calculated probabilities to select the `m` nodes to connect to.

Task 2: Measure and Visualize the Degree Distribution

1. Plot the **aggregate degree distribution** P(k) (the probability that a node has degree k for a B^2 network with `N=10000`, `m=2`, and uniform fitness in `[0,1]`. Use log-log scales.

2. **Compare with the BA model.** On the same plot, add the degree distribution from a BA network with the same `N` and `m`. You can use `nx.barabasi_albert_graph(N, m)`.

3. **Question:** What is the primary visual difference? The B^2 model often has a "flatter" tail, indicating a higher probability of very high-degree nodes (hubs) due to the fitness effect.

Task 3: Fitness-Dependent Degree Distribution - The Key Measurement

The most important test of the B^2 model is to check if `k(η, t)` (the degree of a node with fitness η at time t follows the predicted power-law growth.

1. **Strategy:** For a large network (e.g., `N=5000`), bin your nodes by their fitness. For nodes in a specific fitness bin (e.g., η ~ 0.9, η ~ 0.5), plot their degree against their age (or time of addition).

2. **Plot:** Create a log-log plot of `k_i(t)` vs `t` for different fitness bins.

3. **Expected Result:** You should observe that `k_η(t) ~ t^β(η)`, where the exponent `β(η) = η / C` and `C` is a constant. **Nodes with higher fitness grow**

their degrees faster. This is the "rich-get-richer" effect amplified by intrinsic fitness.

Task 4: Demonstrating the Bose-Einstein Condensation

The condensation occurs when a single node (or a few nodes) grab a macroscopic fraction of the links. This happens when the fitness distribution is "broad enough," e.g., a power-law distribution $\rho(\eta) \sim \eta^{-2}$.

1. **Simulate with a Broad Fitness Distribution.**
    - Implement a power-law fitness distribution, e.g., $\eta = 1 - x$, where x is drawn from a power-law $P(x) \sim x^{-2}$. This will create many low-fitness nodes and a few very high-fitness ones.
    - Generate a B^2 network with $N=2000$, $m=1$, and this new power-law fitness distribution.

2. **Measure the Condensate.**
    - Plot the degree of each node sorted from highest to lowest.
    - Calculate the fraction of total links held by the top 3 nodes.
    - In a non-condensed phase (like uniform fitness), the top nodes have a lot of links, but not a *finite fraction*. In the condensed phase, you will see one node's degree is an order of magnitude larger than the second, and its share of links does not decrease as N grows.